

Phishing Email Detection & Awareness Report

Future Interns – Cyber Security Task 2 (2026)

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Assessment Type

Phishing Email Analysis & Awareness Report

Objective

The objective of this assessment is to analyze phishing email samples, identify phishing indicators, classify email risks, and create awareness guidelines to help users recognize and avoid phishing attacks.

This assessment focuses on:

- Phishing email analysis
 - Email header inspection
 - Suspicious link detection
 - Risk classification
 - User awareness and prevention
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Scope of Analysis

Included

- Email content analysis
- Sender verification
- Header inspection
- Link analysis
- Risk classification
- Awareness recommendations

Excluded

- Malware execution
 - Exploitation
 - Credential testing
 - Live phishing interaction
 - Harmful activities
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Tools Used

- Google Message Header Analyzer
 - MXToolbox Email Header Analyzer
 - Browser URL Inspection
 - Google Docs / PDF Documentation
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Sample Phishing Email 1

⚠ Urgent: Your Account Will Be Locked ⚠ Urgent: Your Account Will Be Locked

Dear User,

We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

👉 Verify Now:
<http://secure-account-verify.com>

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

Analysis – Email Sample 1

Field	Analysis
Sender	support@secure-account-verify.com
Risk Classification	PHISHING
Greeting	Generic greeting (“Dear User”)
Tone	Fear and urgency-based language
Link	Suspicious non-official domain
Threat	Account suspension warning

Phishing Indicators Identified

1. Generic Greeting

The email does not mention the recipient's real name.

2. Urgency & Fear Tactics

The attacker pressures the user to act immediately.

3. Suspicious URL

The domain is not associated with any trusted company.

4. Account Threat

The email threatens account suspension to create panic.

Header Analysis – Email Sample 1

Header Field	Observation
SPF	Failed
DKIM	Missing
Return Path	Different from sender domain
IP Reputation	Suspicious

Risk Explanation

This email is classified as **PHISHING** because it attempts to trick users into revealing sensitive information through fear-based manipulation and fake verification links.

Sample Phishing Email 2

Microsoft Password Expiration NoticeMicrosoft Password Expiration Notice

Dear Employee,

Your Microsoft Outlook password expires today.
Please update your password immediately to avoid losing email access.

Update Password:
<http://microsoft-passwordreset-security.com>

IT Help Desk

Analysis – Email Sample 2

Field	Analysis
Sender	admin@microsoft-passwordreset-security.com
Risk Classification	PHISHING
Brand Abuse	Uses Microsoft name falsely
Link	Fake password reset domain
Urgency	Immediate action requested

Phishing Indicators Identified

1. Fake Microsoft Domain

The domain is not an official Microsoft domain.

2. Password Reset Scam

Attackers attempt to steal login credentials.

3. Urgent Language

The email creates pressure for immediate action.

4. Social Engineering

The attacker pretends to be internal IT support.

Header Analysis – Email Sample 2

Header Field	Observation
SPF	SoftFail
DKIM	Failed
Sender IP	Unknown external server
Reply-To	Different domain detected

Risk Explanation

This email is classified as **PHISHING** because it impersonates Microsoft services to steal employee credentials through a fake password reset page.

Sample Suspicious Email 3

Invoice Attached – Payment PendingInvoice Attached – Payment Pending

Hello,

Please find the attached invoice for pending payment.

Attachment: Invoice_Payment.zip

Finance Department

Analysis – Email Sample 3

Field	Analysis
Sender	finance@unknownbillingservice.net
Risk Classification	SUSPICIOUS
Attachment	ZIP file attachment
Domain	Unknown billing domain
Content	Minimal and unclear

Phishing Indicators Identified

1. Suspicious Attachment

ZIP attachments may contain malware.

2. Unknown Sender

The sender domain is unfamiliar.

3. Lack of Context

No invoice details or company information provided.

4. Generic Message

No personalization or business reference.

Header Analysis – Email Sample 3

Header Field	Observation
SPF	Neutral
DKIM	Missing
Attachment Type	Compressed ZIP file
Mail Server	Unverified server

Risk Explanation

This email is classified as **SUSPICIOUS** because it contains an unexpected compressed attachment from an unknown sender, which may potentially deliver malware.

Risk Classification Summary

Email Sample	Classification
Account Verification Email	PHISHING
Microsoft Password Reset Email	PHISHING
Invoice Attachment Email	SUSPICIOUS

Common Phishing Indicators Observed

- Fake sender domains
- Fear-based language
- Suspicious links
- Generic greetings
- Fake login pages
- Unexpected attachments
- Urgent action requests

- Brand impersonation
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Prevention & Awareness Guidelines

Do's

- Verify sender email addresses
 - Hover over links before clicking
 - Enable Multi-Factor Authentication (MFA)
 - Report suspicious emails immediately
 - Verify requests through official channels
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Don'ts

- Do not click unknown links
 - Do not share passwords or OTPs
 - Do not open suspicious attachments
 - Do not trust urgent threats blindly
 - Do not enter credentials on unknown websites
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Employee Awareness Recommendations

Organizations should:

- Conduct phishing awareness training
 - Simulate phishing campaigns
 - Enforce strong email filtering
 - Implement MFA across systems
 - Encourage incident reporting culture
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Conclusion

Phishing attacks primarily target human behavior rather than technical weaknesses. Attackers use urgency, fear, fake domains, and social engineering techniques to trick users into revealing sensitive information.

This assessment identified multiple phishing indicators across analyzed email samples, demonstrating how users can become targets of credential theft, malware delivery, and account compromise.

Continuous awareness training, careful email verification, and strong security practices are essential to reduce phishing risks within organizations.